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Safety data sheet

BASF safety data sheet. This is a translation of the country-specific safety data sheet into a language other than that required by law. This document does not replace the safety data sheet provided according to Regulation (EC) No 1907/2006.

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Product: **Basonat® HI 190 B/S NG CN**

(ID no. 30675012/SDS_GEN_DE/EN)

Date of print 07.07.2025

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Basonat® HI 190 B/S NG CN

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Chemical
Uses advised against: All consumer uses are strongly advised against., The hazardous properties of the substance require safety measures which can, in principle, not be sufficiently ensured in the home worker sector.

For the detailed identified uses of the product see appendix of the safety data sheet.

1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Global Business Unit Resins and Additives

Telephone: +49 621 60-0
E-mail address: ed-psr@basf.com

1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

For the classification of the mixture the following methods have been applied: extrapolation on the concentration levels of the hazardous substances, on basis of test results and after evaluation of experts. The methodologies used are mentioned at the respective test results.

According to Regulation (EC) No 1272/2008 [CLP]

Flam. Liq. 3	H226 Flammable liquid and vapour.
Acute Tox. 4 (Inhalation - mist)	H332 Harmful if inhaled.
Skin Sens. 1	H317 May cause an allergic skin reaction.
STOT SE 3	H335 May cause respiratory irritation.
Aquatic Chronic 3	H412 Harmful to aquatic life with long lasting effects.

For the classifications not written out in full in this section the full text can be found in section 16.

2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

Pictogram:

Signal Word:

Warning

Hazard Statement:

H226	Flammable liquid and vapour.
H332	Harmful if inhaled.
H317	May cause an allergic skin reaction.
H335	May cause respiratory irritation.
H412	Harmful to aquatic life with long lasting effects.

Precautionary Statements (Prevention):

P280	Wear protective gloves and eye protection or face protection.
P271	Use only outdoors or in a well-ventilated area.
P260	Do not breathe mist or vapour.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.

Precautionary Statements (Storage):

P233	Keep container tightly closed.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Labeling of special preparations (GHS):

EUH204: Contains isocyanates. May produce an allergic reaction. Product contains the following components and may cause an allergic skin reaction: hexamethylene-di-isocyanate
As from 24 August 2023 adequate training is required before industrial or professional use.

Hazard determining component(s) for labelling: HDI oligomers, isocyanurate, hexamethylene-di-isocyanate

2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

No specific dangers known, if the regulations/notes for storage and handling are considered. Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605. The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

SECTION 3: Composition/Information on Ingredients

3.1. Substances

Not applicable

3.2. Mixtures

Chemical nature

HDI oligomers, isocyanurateca. 90 %

dissolved in:Hydrocarbons, C9, aromatics, n-butyl acetate(1:1)

Regulatory relevant ingredients

HDI oligomers, isocyanurate

Content (W/W): >= 75 % - <= 100 %	Acute Tox. 4 (Inhalation - mist)
CAS Number: 28182-81-2	Skin Sens. 1
REACH registration number: 01-2119485796-17	STOT SE 3 (irr. to respiratory syst.)
	H332, H317, H335

n-butyl acetate

Content (W/W): >= 5 % - < 7 % CAS Number: 123-86-4 EC-Number: 204-658-1 REACH registration number: 01-2119485493-29 INDEX-Number: 607-025-00-1	Flam. Liq. 3 STOT SE 3 (drowsiness and dizziness) H226, H336 EUH066

Hydrocarbons, C9, aromatics

Content (W/W): >= 5 % - < 7 % CAS Number: 64742-95-6 REACH registration number: 01-2119455851-35	Asp. Tox. 1 Flam. Liq. 3 STOT SE 3 (drowsiness and dizziness) STOT SE 3 (irr. to respiratory syst.) Aquatic Chronic 2 H226, H304, H336, H335, H411 EUH066

hexamethylene-di-isocyanate

Content (W/W): >= 0,2 % - < 0,3 % CAS Number: 822-06-0 EC-Number: 212-485-8 REACH registration number: 01-2119457571-37 INDEX-Number: 615-011-00-1	Acute Tox. 4 (oral) Acute Tox. 1 (Inhalation - mist) Skin Corr./Irrit. 2 Eye Dam./Irrit. 2 Resp. Sens. 1 Skin Sens. 1 STOT SE 3 (irr. to respiratory syst.) H319, H315, H330, H302, H334, H317, H335 Specific concentration limit: Skin Sens. 1: >= 0,5 % Resp. Sens. 1: >= 0,5 %

Amongst the beforementioned constituents are impurities, stabilizers, or other individual constituents, which are not themselves substances according to Art. 3(1) of Regulation (EC) 1907/2006 (REACH) but belong to other substances and are covered by those. For those constituents, legal information like e.g. identifiers according to Art. 18(2) of Regulation (EC) 1272/2008 (CLP), a chemical safety assessment according to Art 14 of Regulation (EC) 1907/2006 (REACH), or DNEL/PNEC values is not always available or applicable.

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

Immediately remove contaminated clothing.

If inhaled:

If difficulties occur after vapour/aerosol has been inhaled, remove to fresh air and seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Rinse mouth and then drink 200-300 ml of water. Do not induce vomiting unless told to by a poison control center or doctor.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

SECTION 5: Fire-Fighting Measures

5.1. Extinguishing media

Suitable extinguishing media:
dry powder, foam

5.2. Special hazards arising from the substance or mixture

Endangering substances: harmful vapours
Advice: Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

5.3. Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:
The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

SECTION 6: Accidental Release Measures

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Breathing protection required.

6.2. Environmental precautions

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

6.3. Methods and material for containment and cleaning up

For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling

Avoid handling of the substance in case of known skin complaints, hypersensitivity reactions, chronic respiratory disease, asthmatic attacks or bronchial attacks. Avoid contact with the skin, eyes and clothing. Avoid inhalation of dusts/mists/vapours. Avoid frequent and direct contact with substance. Ensure good work practices are implemented. Regular inspection and maintenance of equipment and machines. Clean equipment and the work area every day. Engineering controls have to be used to reduce exposures. Minimise exposure by partial enclosure of the operation or equipment and provide extract ventilation at openings. Workers should receive a pre-placement examination and subsequent periodic medical examinations, including a pulmonary function test. Wear suitable face shield Wear an impervious suit. Use suitable eye protection. Wear chemically resistant gloves in combination with ‘basic’ employee training. Change gloves, if duration of activity exceeds break through time Use a local exhaust ventilation with adequate effectiveness. In case of insufficient ventilation, wear suitable respiratory equipment.

Protection against fire and explosion:
Sources of ignition should be kept well clear. Take precautionary measures against static discharges. If delivered in plastic packing, highest permissable emptying temperature is 5 Kelvin below the flash point.

7.2. Conditions for safe storage, including any incompatibilities

Further information on storage conditions: Keep container tightly closed and in a cool place.

Storage class according to TRGS 510 (originally VCI, Germany): (3) Flammable liquids

Storage stability:
If moisture enters isocyanate containers, CO2 forms and pressure builds up.

7.3. Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

The surveillance of the workplace by exposure measurements may be necessary, in order to prove the efficiency of safety measures, for example ventilation or the need of respiratory protection. Since this requires a specific competency, only accredited laboratories should be contracted. Regarding suitable methods to assess inhalation exposure, the European Standards EN 482, 689 and 14042 are to be considered. In addition, the TRGS 402 has to be observed in Germany.

123-86-4: n-butyl acetate
Short Term Exposure Classification: (TRGS 900 (DE))
Category II: Substances with a resorptive effect
OEL 300 mg/m3 ; 62 ppm (TRGS 900 (DE))
Ceiling limit value/factor: 2
If the occupational exposure limit value (AGW) and the biological limit value (BGW) are complied with, there should be no risk of damage for the unborn child (see TRGS 900, Number 2.7)
STEL value 723 mg/m3 ; 150 ppm (OEL (EU))
indicative
TWA value 241 mg/m3 ; 50 ppm (OEL (EU))
indicative

822-06-0: hexamethylene-di-isocyanate
Short Term Exposure Classification: (TRGS 900 (DE)), Vapor and aerosol
Category I: Substances for which the localized effect has an assigned exposure limit or for substances with a sensitizing effect in respiratory passages
Short Term Exposure Factor: (TRGS 900 (DE)), Vapor and aerosol
Ceiling limit value/factor: 1
Substance listed with exceeding factor and category of short time value.
OEL 0,035 mg/m3 ; 0,005 ppm (TRGS 900 (DE)), Vapor and aerosol
Ceiling limit value/factor: 2

Components with biological limit values

822-06-0: hexamethylene-di-isocyanate
TRGS 903 (DE)
Determinant: Hexamethylenediamine (with hydrolysis)
Biological Specimen: Creatinine in urine
Sampling time: End of shift
Concentration: 15 µg/g Creatinine
TRGS 903 (DE)

Determinant: Hexamethylenediamine (with hydrolysis)
Biological Specimen: Creatinine in urine
Sampling time: End of shift
Concentration: 15 µg/g Creatinine

PNEC

Data refer to the lead substance

Components with PNEC

Hydrocarbons, C9, aromatics
water: 0,1 mg/l
water: 0,1 mg/l
STP: 1,6 mg/l
STP: 1,6 mg/l
soil: 1,1 mg/kg
soil: 1,1 mg/kg
sediment (freshwater): 2,5 mg/kg
sediment (freshwater): 2,5 mg/kg

123-86-4: n-butyl acetate
freshwater: 0,18 mg/l
marine water: 0,018 mg/l
intermittent release: 0,36 mg/l
STP: 35,6 mg/l
sediment (freshwater): 0,981 mg/kg
sediment (marine water): 0,0981 mg/kg
soil: 0,0903 mg/kg

28182-81-2: HDI oligomers, isocyanurate
freshwater: 0,127 mg/l
marine water: 0,0127 mg/l
intermittent release: 1,27 mg/l
sediment (freshwater): 266700 mg/kg
marine water: 26670 mg/kg
soil: 53182 mg/kg
STP: 38,28 mg/l

DNEL

Data refer to the lead substance

Components with DNEL

28182-81-2: HDI oligomers, isocyanurate
worker: Short-term exposure - local effects, Inhalation: 1 mg/m3
worker: Long-term exposure - local effects, Inhalation: 0,5 mg/m3

123-86-4: n-butyl acetate

worker: Short-term exposure - local effects, Inhalation: 600 mg/m3
worker: Long-term exposure - local effects, Inhalation: 300 mg/m3
consumer: Short-term exposure - local effects, Inhalation: 300 mg/m3
consumer: Long-term exposure - local effects, Inhalation: 35,7 mg/m3
worker: Long-term exposure- systemic effects, dermal: 11 mg/kg
worker: Short-term exposure - systemic effects, dermal: 11 mg/kg
consumer: Long-term exposure- systemic effects, dermal: 6 mg/kg
consumer: Short-term exposure - systemic effects, dermal: 6 mg/kg
consumer: Long-term exposure- systemic effects, oral: 2 mg/kg
consumer: Short-term exposure - systemic effects, oral: 2 mg/kg

Hydrocarbons, C9, aromatics
worker: Long-term exposure- systemic effects, by inhalation: 150 mg/m3
worker: Long-term exposure- systemic effects, by inhalation: 150 mg/m3
worker: Long-term exposure- systemic effects, dermal: 25 mg/kg
worker: Long-term exposure- systemic effects, dermal: 25 mg/kg
consumer: Long-term exposure- systemic effects, by inhalation: 32 mg/m3
consumer: Long-term exposure- systemic effects, by inhalation: 32 mg/m3
consumer: Long-term exposure- systemic effects, dermal: 11 mg/kg
consumer: Long-term exposure- systemic effects, dermal: 11 mg/kg
consumer: Long-term exposure- systemic effects, oral: 11 mg/kg
consumer: Long-term exposure- systemic effects, oral: 11 mg/kg

8.2. Exposure controls

Personal protective equipment

Respiratory protection:
For short-term or slight exposure, use a respiratory mask with filter, for intensive or long-term exposure wear full respiratory protection apparatus Combination filter for gases/vapours of organic compounds and solid and liquid particles (f.e. EN 14387 Type A-P2)

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials for short-term contact (recommended: At least protective index 2, corresponding > 30 minutes of permeation time according to EN ISO 374-1)
nitrile rubber (NBR) - 0.4 mm coating thickness
fluoroelastomer (FKM) - 0.7 mm coating thickness
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing. Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Safety glasses with side-shields (frame goggles) (f.e. EN 166) and face shield

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is required additionally to the stated personal protection equipment. Do not breathe vapour/aerosol/spray mists. Ensure adequate ventilation. Clean equipment, work area and clothing regularly.

SECTION 9: Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	liquid	
Form:	liquid	
Colour:	colourless to slightly yellow	
Odour:	faint odour	
Odour threshold:	not determined	
Melting point:	-25 °C	
Boiling point:	160 °C	
Flammability:	Flammable liquid and vapour.	(derived from flash point)
Lower explosion limit:	1,0 %(V)	
Upper explosion limit:	7,5 %(V)	
Flash point:	51,5 °C	(ISO 13736)
Auto-ignition temperature:	430 °C	(DIN 51794)
Thermal decomposition:	No decomposition if used correctly.	
pH value:	not determined	
Viscosity, dynamic:	450 - 650 mPa.s (23 °C)	(DIN 53019)
Thixotropy:	not thixotropic	
Solubility in water:	Reacts with water.	
Partitioning coefficient n-octanol/water (log Kow):	not applicable	
Vapour pressure:	not determined	
Relative density:	1,13 (20 °C)	
Density:	1,13 g/cm3 (20 °C)	(DIN 51757)
Relative vapour density (air):	not determined	

Particle characteristics

Particle size distribution:	The substance / product is marketed or used in a non solid or granular form. -

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: not explosive

Oxidizing properties

Fire promoting properties: not fire-propagating

Other safety characteristics

Miscibility with water: Reacts with water.

Hygroscopy: Non-hygroscopic

Surface tension: Based on chemical structure, surface activity is not to be expected.

Evaporation rate: not determined

SECTION 10: Stability and Reactivity

10.1. Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

10.2. Chemical stability

The product is chemically stable.

10.3. Possibility of hazardous reactions

Reacts with alcohols. Reacts with amines. Reacts with substances which contain active hydrogen. Reacts with water, with formation of carbon dioxide. The formation of gaseous decomposition products builds up pressure in tightly closed containers. Vapours may form ignitable mixture with air.

10.4. Conditions to avoid

Avoid moisture. See SDS section 7 - Handling and storage.

10.5. Incompatible materials

Substances to avoid:
water, alcohols, amines

10.6. Hazardous decomposition products

No hazardous decomposition products if stored and handled as prescribed/indicated.

SECTION 11: Toxicological Information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Assessment of acute toxicity:
Virtually nontoxic after a single ingestion. Virtually nontoxic after a single skin contact. May be harmful if inhaled. The product has not been tested. The statement has been derived from the properties of the individual components.

Experimental/calculated data:
ATE (oral): > 5.000 mg/kg (calculated)

ATE (dermal): > 5.000 mg/kg (calculated)
The product has not been tested. The statement has been derived from the properties of the individual components.

Irritation

Assessment of irritating effects:
Not irritating to eyes and skin.

Experimental/calculated data:
Skin corrosion/irritation
rabbit: non-irritant (OECD Guideline 404)
The product has not been tested. The statement has been derived from the properties of the individual components.

Serious eye damage/irritation
rabbit: non-irritant (OECD Guideline 405)
The product has not been tested. The statement has been derived from the properties of the individual components.

Respiratory/Skin sensitization

Assessment of sensitization:
Sensitization after skin contact possible.

Experimental/calculated data:
Guinea pig maximization test guinea pig: skin sensitizing
The product has not been tested. The statement has been derived from the properties of the individual components.

Germ cell mutagenicity

Assessment of mutagenicity:
Based on the ingredients, there is no suspicion of a mutagenic effect.

Carcinogenicity

Assessment of carcinogenicity:
The whole of the information assessable provides no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:
Based on the ingredients, there is no suspicion of a toxic effect on reproduction.

Developmental toxicity

Assessment of teratogenicity:
Based on the ingredients, there is no suspicion of a teratogenic effect.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Causes temporary irritation of the respiratory tract.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The information available on the product provides no indication of toxicity on target organs after repeated exposure.

Aspiration hazard

No aspiration hazard expected.

Interactive effects

No data available.

11.2. Information on other hazards

Endocrine disrupting properties

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

SECTION 12: Ecological Information

12.1. Toxicity

Assessment of aquatic toxicity:
Harmful to aquatic life. Harmful to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from the properties of the individual components.

Toxicity to fish:
LC50 (96 h) 10 - 100 mg/l, Brachydanio rerio
The product has not been tested. The statement has been derived from the properties of the individual components.

Aquatic invertebrates:
EC50 10 - 100 mg/l, Daphnia magna
The product has not been tested. The statement has been derived from the properties of the individual components.

Aquatic plants:
EC50 (72 h) 10 - 100 mg/l, algae
The product has not been tested. The statement has been derived from the properties of the individual components.

Microorganisms/Effect on activated sludge:
EC50 (3 h) > 1.000 mg/l, bacteria
The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations. The product has not been tested. The statement has been derived from the properties of the individual components.

Chronic toxicity to fish:
No data available.

Chronic toxicity to aquatic invertebrates:

No data available.

Assessment of terrestrial toxicity:
No data available concerning terrestrial toxicity.

12.2. Persistence and degradability

Elimination information:
Not readily biodegradable (by OECD criteria).

12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested.

12.4. Mobility in soil

Assessment transport between environmental compartments:
Volatility: No data available.

12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

12.6. Endocrine disrupting properties

Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

12.7. Other adverse effects

The product does not contain substances that are listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

12.8. Additional information

Add. remarks environm. fate & pathway:
Treatment in biological waste water treatment plants has to be performed according to local and administrative regulations.

Other ecotoxicological advice:
Do not release untreated into natural waters. The local regulations on waste-water treatment must be followed. Do not discharge product into the environment without control. The product has not been tested. The statement has been derived from the properties of the individual components.

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:
Uncontaminated packaging can be re-used.
Packs that cannot be cleaned should be disposed of in the same manner as the contents.

SECTION 14: Transport Information

Land transport

ADR

UN number or ID number:	UN1866
UN proper shipping name:	RESIN SOLUTION
Transport hazard class(es):	3
Packing group:	III
Environmental hazards:	no
Special precautions for user:	Tunnel code: D/E

RID

UN number or ID number:	UN1866
UN proper shipping name:	RESIN SOLUTION
Transport hazard class(es):	3
Packing group:	III
Environmental hazards:	no
Special precautions for user:	None known

Inland waterway transport
ADN

UN number or ID number:	UN1866
UN proper shipping name:	RESIN SOLUTION
Transport hazard class(es):	3
Packing group:	III
Environmental hazards:	no
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

UN number or ID number:	UN 1866
UN proper shipping name:	RESIN SOLUTION
Transport hazard class(es):	3
Packing group:	III
Environmental hazards:	no Marine pollutant: NO
Special precautions for user:	EmS: F-E; S-E

Air transport

IATA/ICAO

UN number or ID number:	UN 1866
UN proper shipping name:	RESIN SOLUTION
Transport hazard class(es):	3
Packing group:	III
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for user:	None known

14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

SECTION 15: Regulatory Information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Prohibitions, Restrictions and Authorizations

Annex XVII of Regulation (EC) No 1907/2006: Number on List: 40, 3, 74, 75

Hazardous Incident Ordinance (Germany):
List entry in regulation: 1.2.5.3

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
List entry in regulation: H2
List entry in regulation: P5c

Water hazard class (§8/§10 AwSV (Self-classification of the mixture according to calculation method)): (2) significantly water polluting.

Law on the Protection of Working Youth
German Regulation TA Luft (Technical Instruction on Air Quality Control, i.e. first Directive to the Federal Immission Control Ordinance)

The specifications of the Technical Rule for Hazardous Substances (TRGS) 401 must be observed (TRGS 401: Risks resulting from skin contact - identification, assessment, measures).
German Ordinance on the Curbing of Emissions of Volatile Organic Compounds when using Organic Solvents (31. BImSchV), see also 2010/75/EU
Second ordinance on the implementation of the Federal Immission Control Act (Ordinance to limit the emission of volatile halogenated organic compounds)
TRGS 401: 'Risks resulting from skin contact - identification, assessment, measures' and TRBA/TRGS 406: 'compounds sensitizing to the respiratory system'
Regulation on prohibitions and restrictions on the marketing of dangerous substances, preparations and goods in accordance with the chemical law (Germany)
The Maternity Protection Act needs to be considered.
TRGS 430 'Isocyantates - risk assessment and safety precautions' (Germany)

15.2. Chemical Safety Assessment

Assessment of safe use has been performed for the mixture and the result is attached as an annex to the SDS

SECTION 16: Other Information

The product is used mainly as a hardener in coating materials or adhesives. The handling of coating materials or adhesives containing reactive polyisocyanates and residual monomeric HDI requires appropriate protective measures referred to in this safety data sheet. These products may therefore be used only in industrial or trade applications. They are not suitable for use in homeworker (DIY) applications.

Flam. Liq.	Flammable liquids
Acute Tox.	Acute toxicity
Skin Sens.	Skin sensitization
STOT SE	Specific target organ toxicity — single exposure
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Asp. Tox.	Aspiration hazard
Skin Corr./Irrit.	Skin corrosion/irritation
Eye Dam./Irrit.	Serious eye damage/eye irritation
Resp. Sens.	Respiratory sensitization
H226	Flammable liquid and vapour.
H332	Harmful if inhaled.
H317	May cause an allergic skin reaction.
H335	May cause respiratory irritation.
H412	Harmful to aquatic life with long lasting effects.
H336	May cause drowsiness or dizziness.
H304	May be fatal if swallowed and enters airways.
H411	Toxic to aquatic life with long lasting effects.
H319	Causes serious eye irritation.
H315	Causes skin irritation.
H330	Fatal if inhaled.
H302	Harmful if swallowed.
H334	May cause allergy or asthma symptoms or breathing difficulties if inhaled.
EUH066	Repeated exposure may cause skin dryness or cracking.

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Annex: Exposure Scenarios

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1. Short title of exposure scenario

Formulation
SU3, SU10; ERC2; PROC1, PROC2, PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC2: Formulation into mixture ESVOC SpERC 2.2.v1: ESVOC SpERC 2.2.v1
Operational conditions	
Minimum emission days per year Continuous	
Emission factor air	2,5 %
Emission factor water	0,2 %
Emission factor soil	0,01 %
	Releases based on ESVOC/CEFIC information
Dilution factor river	10
Dilution factor coast	100
Risk Management Measures	
Treat air emissions to provide a typical removal efficiency of (%)	56,5 %
Treat wastewater (prior to discharge to STP) to provide the required removal efficiency of (%)	94,7 %
	Prevent discharge of undissolved substance to or recover from wastewater

Type of STP	Municipal STP
Estimated subst. removal from wastewater via sewage treatm. (%)	95,5 %
Total effic. of removal from wastewater after RMMs and STP(%)	95,5 %
Assumed sewage treatment plant flow (m3/d)	2.000 m3/d
Sludge Treatment	Do not use sludge as fertiliser
Exposure estimate and reference to its source	
Risk Characterization Ratio (RCR)	0,98
	Risk from environmental exposure is driven by humans via indirect exposure (primarily inhalation).
Maximum amount of safe use	1.000 t/d
Risk from environmental exposure is driven by humans via indirect exposure (primarily inhalation).	

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2. Short title of exposure scenario

Industrial applications
SU3, SU12, SU13, SU19; ERC4; PROC1, PROC2, PROC3, PROC4, PROC5, PROC7, PROC8a, PROC8b, PROC9, PROC10, PROC13, PROC14, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ESVOC SpERC 4.3a.v1: ESVOC SpERC 4.3a.v1
Operational conditions	
Emission factor air	98 %
Emission factor water	0,007 %
Emission factor soil	0 %
Receive Surf. Water (Flow Rate).	18.000 m3/d
Dilution factor river	10
Dilution factor coast	100
Risk Management Measures	
Air treatment measures considered suitable are, e.g.	Wet scrubber - for dusts, Filtration, Waste gas treatment by thermal oxidation, Adsorption
Wastewater treatment measures considered suitable are, e.g.	Acclimated biological treatment, Distillation
Type of STP	Municipal STP

Assumed sewage treatment plant flow (m3/d)	2.000 m3/d
Exposure estimate and reference to its source	
Assessment method	EASY TRA v4.2, ECETOC TRA v3.0, Environment
Risk Characterization Ratio (RCR)	0,0024
	Risk from environmental exposure is driven by freshwater.
Maximum amount of safe use	880 t/d
Risk from environmental exposure is driven by freshwater.	

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3. Short title of exposure scenario

Professional applications
SU22, SU10, SU12, SU13, SU19; ERC2, ERC8a, ERC8b, ERC8d, ERC8e; PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC10, PROC11, PROC13, PROC14, PROC15

Control of exposure and risk management measures

Contributing exposure scenario	
Use descriptors covered	ERC2: Formulation into mixture ESVOC SpERC 2.2.v1: ESVOC SpERC 2.2.v1
Operational conditions	
Minimum emission days per year Continuous	
Emission factor air	2,5 %
Emission factor water	0,2 %
Emission factor soil	0,01 %
	Releases based on ESVOC/CEFIC information
Dilution factor river	10
Dilution factor coast	100
Risk Management Measures	
Treat air emissions to provide a typical removal efficiency of (%)	56,5 %
Treat wastewater (prior to discharge to STP) to provide the required removal efficiency of (%)	94,7 %
	Prevent discharge of undissolved substance to or recover from wastewater
Type of STP	Municipal STP
Estimated subst. removal from wastewater via sewage treatm. (%)	95,5 %
Total effic. of removal from wastewater after RMMs and STP(%)	95,5 %
Assumed sewage treatment plant flow (m3/d)	2.000 m3/d

Sludge Treatment	Do not use sludge as fertiliser
Exposure estimate and reference to its source	
Risk Characterization Ratio (RCR)	0,98
	Risk from environmental exposure is driven by humans via indirect exposure (primarily inhalation).
Maximum amount of safe use	1.000 t/d
Risk from environmental exposure is driven by humans via indirect exposure (primarily inhalation).	

Contributing exposure scenario	
Use descriptors covered	ESVOC SpERC 8.3b.v1: ESVOC SpERC 8.3b.v1
Operational conditions	
Emission factor air	98 %
Emission factor water	1 %
Emission factor soil	1 %
Receive Surf. Water (Flow Rate).	18.000 m3/d
Dilution factor river	10
Dilution factor coast	100
Risk Management Measures	
Type of STP	Municipal STP
Assumed sewage treatment plant flow (m3/d)	2.000 m3/d
Exposure estimate and reference to its source	
Assessment method	EASY TRA v4.2, ECETOC TRA v3.0, Environment
Risk Characterization Ratio (RCR)	0,000001
	Risk from environmental exposure is driven by freshwater.
Maximum amount of safe use	47.000 kg/d
Risk from environmental exposure is driven by freshwater.	

Contributing exposure scenario	
Use descriptors covered	ESVOC SpERC 8.3b.v1: ESVOC SpERC 8.3b.v1
Operational conditions	
Emission factor air	98 %
Emission factor water	1 %
Emission factor soil	1 %

Receive Surf. Water (Flow Rate).	18.000 m3/d
Dilution factor river	10
Dilution factor coast	100
Risk Management Measures	
Type of STP	Municipal STP
Assumed sewage treatment plant flow (m3/d)	2.000 m3/d
Exposure estimate and reference to its source	
Assessment method	EASY TRA v4.2, ECETOC TRA v3.0, Environment
Risk Characterization Ratio (RCR)	0,000001
	Risk from environmental exposure is driven by freshwater.
Maximum amount of safe use	47.000 kg/d
Risk from environmental exposure is driven by freshwater.	

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